

$$\mathbb{Q}(\alpha) \cong \frac{\mathbb{Q}[x]}{\langle x^3 - 3x - 1 \rangle} \text{ iff } \text{irreducible poly.}$$

Let us assume that $x^3 - 3x - 1$ has a rational root.

$$\beta = \frac{p}{q} \Rightarrow \left(\frac{p}{q}\right)^3 - 3\frac{p}{q} - 1 = 0 \quad \checkmark$$

$$\Rightarrow p^3 - 3pq^2 - q^3 = 0 \quad \checkmark$$

$$\Rightarrow q^2 \mid p^3 \text{ but } \gcd(q, p) = 1.$$

$$\Rightarrow p^3 = q^2 k$$

$$\Rightarrow \exists n, y \in \mathbb{Z} \text{ s.t.}$$

$$-pn + qy = 1$$

$$\Rightarrow p^3 n + p^2 q y = p^2$$

$$\Rightarrow q^2 k n + p^2 q y = p^2$$

$$\Rightarrow p^2 \mid q^2$$

$$\Rightarrow q^2 = p^2 k_1$$

$$q p n + q^2 y = q$$

$$\Rightarrow p n + p^2 k_1 = \underline{q}$$

$$\Rightarrow q \mid p^2$$

$$\Rightarrow p^2 = q k_2$$

$$p^2 n + p^2 q y = p$$

$$\Rightarrow q k_2 n + p q y = \underline{p}$$

$$\Rightarrow \boxed{p \mid q} \quad (\Rightarrow \checkmark)$$

$\mathbb{Q}(\alpha)$ is of deg. 3 extension.

(b) $\alpha^4 + 2\alpha^3 + 3 \in \mathbb{Q}(\alpha) = \{a + b\alpha + c\alpha^2 \mid a, b, c \in \mathbb{Q}\}$

$$\begin{array}{r}
 x^3 - 3x - 1 \mid x^4 + 2x^3 + 3(x+2) \\
 \underline{-(x^4 - 3x^2 - x)} \\
 2x^3 + 3x^2 + x + 3 \\
 \underline{-(2x^3 - 6x - 2)} \\
 3x^2 + 7x + 5
 \end{array}$$

(c) $(x-1)^{-1}$ in form $a + bx + cx^2$

$$(a + bx + cx^2)(x-1) \equiv 1 \pmod{(x^3 - 3x - 1)} \quad \checkmark \quad \textcircled{\times} \textcircled{\times}$$

$$\Rightarrow ax + bx^2 + cx^3 - a - bx - cx^2 \equiv 1$$

$$\Rightarrow (a-b)x + (b-c)x^2 + cx^3 - a \equiv 1$$

$$\begin{array}{r}
 \checkmark (x^3 - 3x - 1) \mid cx^3 + (b-c)x^2 + (a-b)x - a \\
 \underline{-(cx^3 - 3cx - c)} \\
 (b-c)x^2 + (a-b+3c)x - a + c \\
 \hline
 \parallel \\
 1
 \end{array}$$

$$b - c = 0$$

$$a - b + 3c = 0 \quad c - a = 1 \Rightarrow c = 1 - \frac{2}{3}$$

$$a + 2c = 0$$

$$= \frac{1}{3}$$

$$a + 2(1+a) = 0$$

$$b = \frac{1}{3}$$

$$\Rightarrow 3a = -2$$

$$\Rightarrow a = -\frac{2}{3}$$

$$\boxed{-\frac{2}{3} + \frac{1}{3}\alpha + \frac{1}{3}\alpha^2 = (\alpha-1)^{-1}}$$